③ Conservative forces, potential

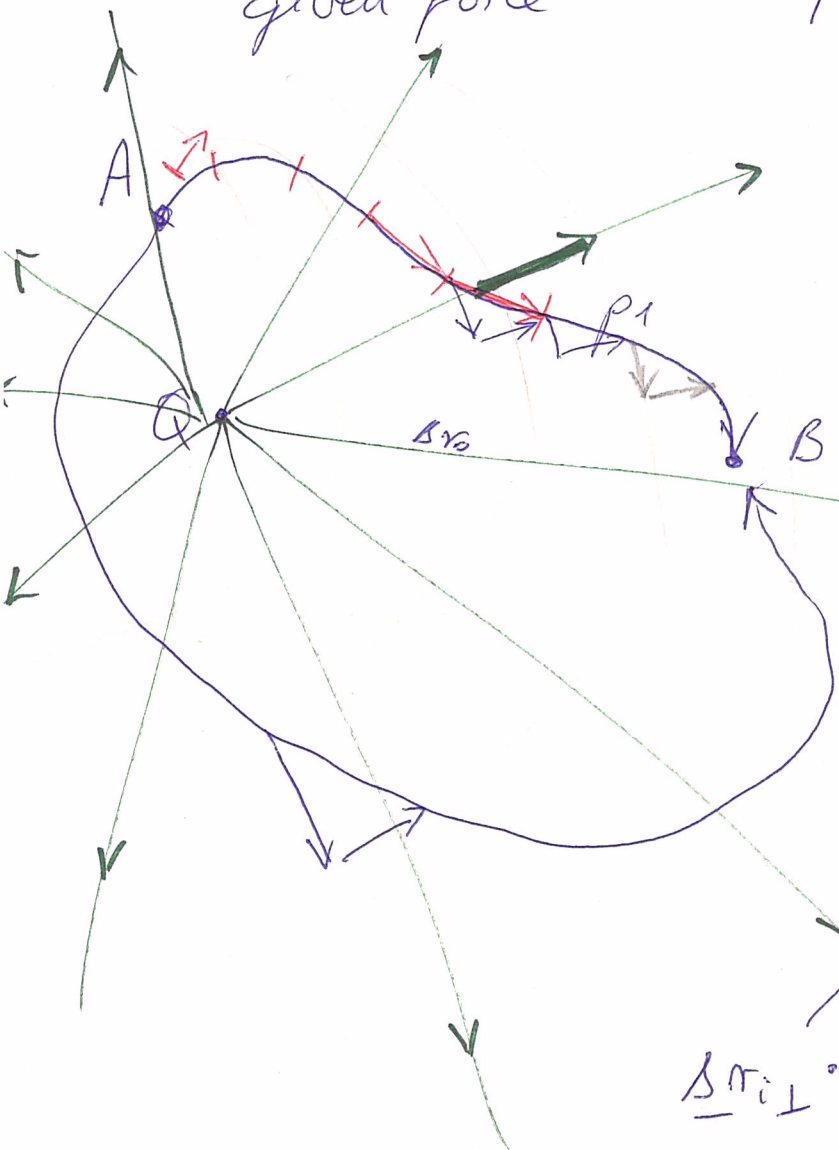
Is electrical field conservative?

Conservative forces: The work made by the given force is independent of the path: (for any 2 points)

$$\Delta W_1 = \Delta \underline{r}_1 \cdot \underline{F}(\underline{r}_1)$$

$$W_{AB} = \lim_{\Delta r_i \to 0} \sum_i \underline{F}_i \Delta \underline{r}_i = \int_A^B \underline{F}\, d\underline{r}$$

(path integral)



Decompose Δr_i into radial and tangential (normal) components ▽ By intersecting of concentric spheres!

$\Delta \underline{r}_{i\perp} \cdot \underline{F}(r_i) = 0$ Work is only made by the radial part!

$\Delta \underline{r}_i = \Delta \underline{r}_{\parallel} + \Delta \underline{r}_{i\perp}$

~~$W_{AB} = \sum \underline{F}_i(r_i) \cdot \Delta \underline{r}_{i\parallel}$~~

$$\Delta W_i^{\rho_1} = \Delta r_{\parallel} \cdot F^{\rho_1}(r_i)$$

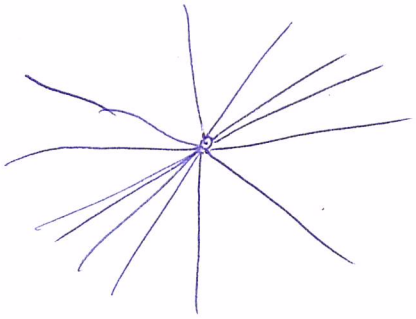
$$\Delta W_i^{\rho_2} = \Delta r_{\parallel} \cdot F^{\rho_2}(r_i)$$

F only depends on the distance

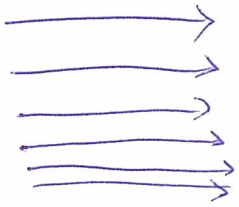
(like all central forces)

$\Delta W_i^{\rho_1} = \Delta W_i^{\rho_2} \Rightarrow$ Conservati

is this conservative :



?



?

Conservative force

↳ we can define potential energy!

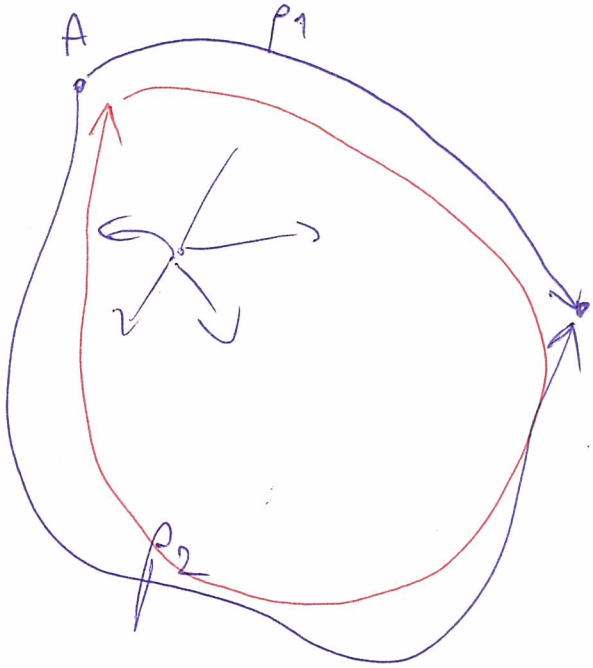
$$E_{pot}^{A}(B) = -W_{AB} = -\int_{A}^{B} \underline{F}\,d\underline{r} = -q\int_{A}^{B} \underline{E}\,d\underline{r}$$

($E_{pot}^{A}(B) < 0$ on the figure)

Potential energy of point B in the field ~~with refer~~ compared to point A as reference point.

(if independent of the path - conservative force - the pot. energy is a unique function of coordinates A and B

Let's integrate in 2 different paths! why?



$$(P_1)\int_{A}^{B} \underline{E}\,d\underline{r} = (P_2)\int_{A}^{B} \underline{E}\,d\underline{r} = -P_2\int_{B}^{A} \underline{E}\,d\underline{r}$$

$$(P_1)\int_{A}^{B} \underline{E}\,d\underline{r} + (P_2)\int_{B}^{A} \underline{E}\,d\underline{r} = 0$$

$$\oint \underline{E}\,d\underline{r} = 0 \text{ for any closed path!}$$

first law of static electricity

expressing the conservativity of static el. field.

~~Electro~~ Static electric field is vortex-free

(whirl, eddy)

$$\oint \underline{E}\,d\underline{r} = \iint \operatorname{rot} \underline{E}\,d\underline{s} = 0$$

$$\Updownarrow$$

$$\operatorname{rot} \underline{E} = 0$$

$\Rightarrow$ ~~$\exists U$~~ / ~~$U^A(B)$~~

~~$\underline{E} = -\operatorname{grad} U$~~

(I)

vortex density is zero

$$E_{pot}^{(0)}(P) = q\left[-\int_0^P \underline{E}\,d\underline{r}\right]$$

↑ test charge

↖ Potential in point P

$$u^{(0)}(P) = -\int_0^P \underline{E}\,d\underline{r}$$

Note: $\Delta E_{pot} = -W = -q\,\Delta U$ applied for an elementary charge e: eV energy

$\Delta E_{pot} = e \cdot \Delta U \Rightarrow 1{,}6 \cdot 10^{-19}$ J

$$[U] = 1\,\frac{N}{C}\cdot m = 1V$$

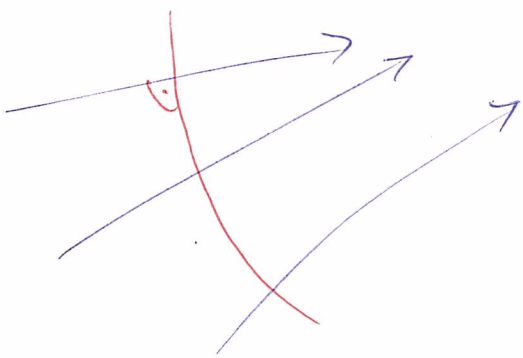
How to reconstruct $\underline{E}$ from U?

$$dU = -\underline{E}\,d\underline{r} \longrightarrow \left\| \frac{dU}{d\underline{r}} \right\| = -\nabla U = -\text{grad}\,U$$

$$\underline{E} = -\text{grad}(U)$$

Equipotential surface:

$U(x,y,z) = \text{const}$. That is by moving on this surface $\underline{E}\,d\underline{r} = 0$



E is always ⊥ to equipot surfaces!

I. $\oint \underline{E}\, d\underline{r} = 0 \Rightarrow \exists U \qquad U^0(P) = -\int_0^P \underline{E}\, d\underline{r}$

$$\underline{E} = -\mathrm{grad}(U)$$

II. $\oint \underline{E}\, d\underline{s} = \frac{Q}{\varepsilon_0} \Leftrightarrow \mathrm{div}\, \underline{E} = \frac{\varrho}{\varepsilon_0}$

$$\mathrm{div}(\mathrm{grad}(U)) = -\frac{\varrho}{\varepsilon_0}$$

$$\Delta U = -\frac{\varrho}{\varepsilon_0}$$

↑ Laplace operator

Poisson equation

We have 2 methods to calculate the field (the potential)

① solving Poisson

② using Coulomb's law + superposition pr.

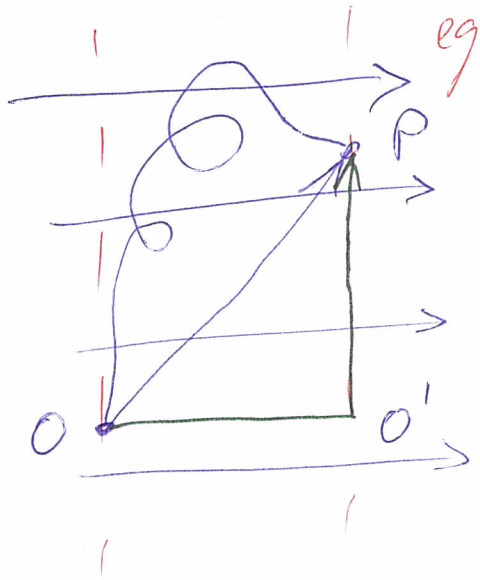
Are we ready? NO

Difficulties:

- Charges move
- We can not accurately determine the position of charges
- computationally catastrophic

↳ Let's investigate ⇄ conductors as macroscopic insulators bodies

Potential ~~difference~~ in a homogeneous field

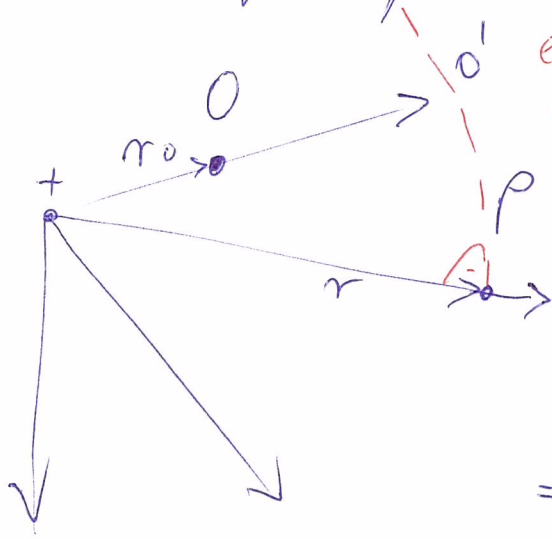


$$U^0(P) = -\int_O^P \underline{E}\,d\underline{r} =$$

$$= -\int_O^{O'} \underline{E}\,d\underline{r} - \int_{O'}^{P} \underline{E}\,d\underline{r} =$$

$$= -E \cdot L$$

Potential of a point charge:



equipot surface: sphere

$$U^0(P) = -\int_O^P \frac{1}{4\pi\varepsilon_0} Q \frac{\underline{r}'}{r'^3} \cdot d\underline{r}' =$$

$$U^0(P) = -\frac{1}{4\pi\varepsilon_0} \cdot Q \int_O^{O'} \frac{\underline{r}'\,d\underline{r}'}{r'^3} + \int_{O'}^{P} \frac{\underline{r}}{r^3}\,d\underline{r}$$

$$= -\frac{1}{4\pi\varepsilon_0} \cdot Q \cdot \int_{r_0}^{r} \frac{dr'}{r'^2} = -\frac{1}{4\pi\varepsilon_0} Q \left[-\frac{1}{r}\right]_{r_0}^{r}$$

$$U^0(P) = \frac{1}{4\pi\varepsilon_0} \cdot Q \left(\frac{1}{r} - \frac{1}{r_0}\right)$$

To make it simpler: let's eliminate $\frac{1}{r_0} \nabla$

reference point is usually moved to ∞ !

$$U^\infty(P) = U(P) = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q}{r}$$

4/b

Cross check: $\underline{E} = -\mathrm{grad}(U(P)) = ?$ HW